

MAX3010x Sensor Module Failure Analysis Report

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Prepared from ESP32/PlatformIO diagnostic testing

Conclusion: Module is incapable of producing valid pulse-oximetry data.

Executive Summary

A MAX3010x-compatible optical sensor module was evaluated using an ESP32 development board and a series of progressively lower-level diagnostic tests. The module responded on the I2C bus at address 0x57 and returned a Part ID value of 0x11. However, all tests involving the measurement engine, FIFO generation, LED drive circuitry, and reset behavior failed. The evidence indicates that the module is either a defective device or a counterfeit/clone that only partially emulates the MAX3010x register interface.

Any IR, Red, SpO2, or heart-rate values obtained from this module must be considered invalid. Further software tuning, register adjustments, library changes, timing changes, pull-up modifications, or filtering techniques cannot compensate for the absence of a functioning optical acquisition subsystem.

Test Environment

Controller: ESP32 Dev Module

Development Environment: VS Code + PlatformIO

Interface: I2C (GPIO21 SDA, GPIO22 SCL)

Supply Voltage: 3.3 V

Sensor Address: 0x57

Sensor Board Marking: MAX30100/MAX30102 compatible module shown aside.



Figure 1 : Oximeter Sensor Board

Diagnostic Procedure

Testing progressed from high-level library validation to direct register access. Each subsystem was isolated and verified independently:

1. I2C bus detection
2. Device identification register read
3. Register read/write verification
4. Soft reset verification

5. Mode configuration verification
6. FIFO pointer monitoring
7. FIFO data acquisition
8. LED emission verification using a phone camera
9. Manual register dump analysis

Results

I2C Scan:

- Device consistently acknowledged at address 0x57.
- ESP32 wiring and bus operation confirmed.

Part ID Test:

- Register 0xFF returned 0x11.
- Device appeared to identify itself as a MAX30105-class device.

Register Read/Write:

- Mode register accepted writes.
- LED current registers accepted writes.
- Several configuration registers failed to behave according to the datasheet.

Soft Reset Test:

- Reset command set the reset bit.
- Reset bit never self-cleared.
- Genuine MAX3010x devices automatically clear this bit after reset completion.

FIFO Operation:

- FIFO write pointer remained at 0.
- FIFO read pointer remained at 0.
- No samples were generated.

Raw Data Acquisition:

- Returned values were predominantly 0.
- Occasional random values appeared.
- Communication errors intermittently occurred.
- No stable physiological signal was observed.

LED Emission Test:

- Red LED not visible.
- IR LED not visible through a smartphone camera.
- Indicates LED driver subsystem never activated.

Register Dump Analysis

Observed:

0x08 = 0x0F

0x09 = 0x03

0x0A = 0x07

0x0C = 0x24

0x0D = 0x24

Interpretation:

- Configuration registers were partially writable.
- The SpO2 configuration register did not fully retain programmed values.
- FIFO pointers remained static.
- Measurement engine never entered normal acquisition operation.

Subsystem Assessment

I2C Interface: PASS

Register Access Logic: PASS (partial)

Internal State Machine: FAIL

Measurement Engine: FAIL

FIFO Generator: FAIL

LED Drivers: FAIL

Optical Front End: FAIL

Pulse Oximetry Function: FAIL

Validity of Output Data

Because the FIFO never fills and the LEDs never emit, the sensor is not collecting optical measurements. Any numerical values read from FIFO registers are therefore not derived from physiological signals.

Consequently:

- Heart rate calculations are invalid.
- SpO2 calculations are invalid.
- Raw IR values are invalid.
- Raw Red values are invalid.
- Signal-processing or filtering cannot recover missing data.

The issue is not signal quality; the issue is the absence of signal generation itself.

Why Further Tuning Will Not Help

Numerous software-level adjustments were already attempted:

- Different I2C clock frequencies
- Direct register access
- Library-based initialization
- Manual configuration
- Soft reset procedures
- FIFO monitoring

These actions can only affect a functioning sensor. The module never generated samples and never activated its LEDs. Since the measurement subsystem is inactive, no amount of software tuning can produce valid pulse-oximetry data.

The limiting factor is hardware functionality, not firmware configuration.

Final Conclusion

The module cannot be considered a functional MAX30102/MAX30105 sensor. Although it responds on I2C and exposes a partially working register map, the measurement subsystem fails all functional tests.

The most likely explanations are:

1. Counterfeit or clone device with incomplete functionality.
2. Factory-defective module.

Replacement of the module is recommended. Continued debugging is unlikely to yield a working pulse-oximeter because the fundamental optical acquisition hardware is non-operational.